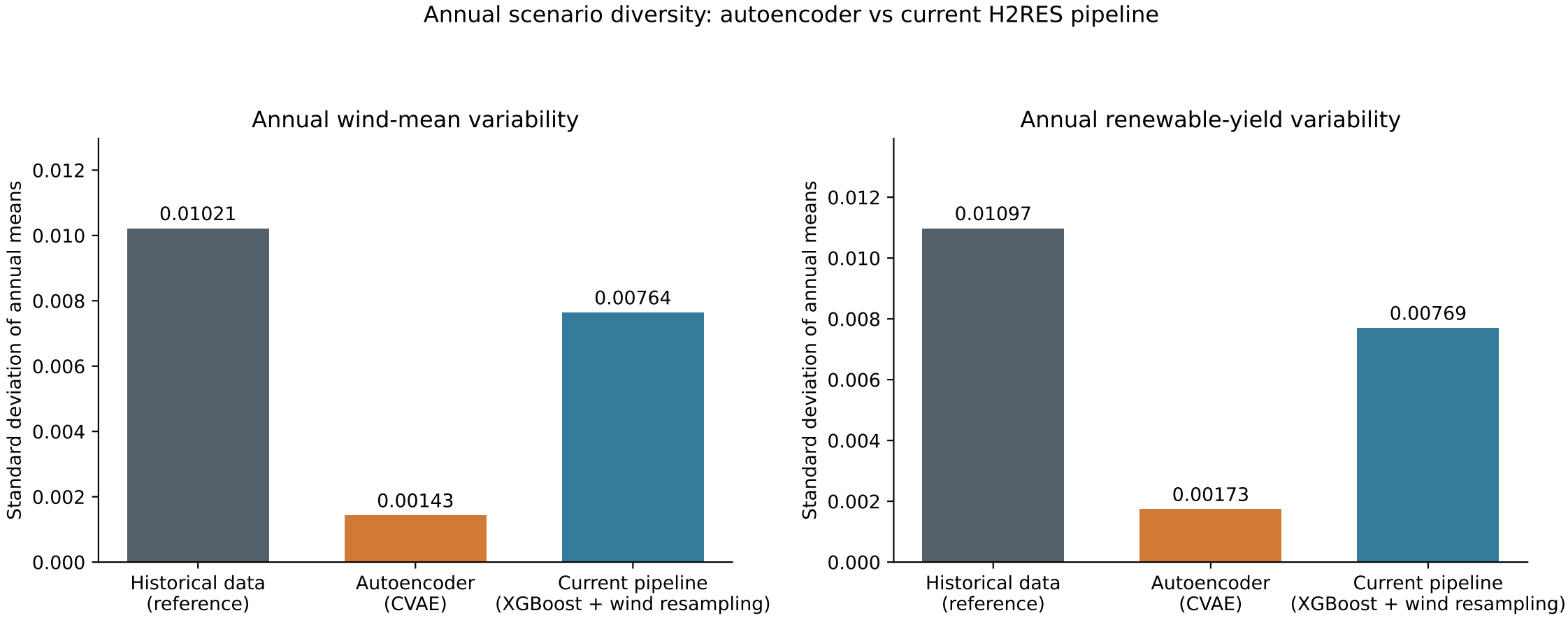
Annual scenario diversity: autoencoder vs current H2RES pipeline
Annual wind-mean variability
Annual renewable-yield variability
Standard deviation of annual means
0.000
0.002
0.004
0.006
0.008
0.010
0.012
0.01021
0.00143
0.00764
0.01097
0.00173
0.00769
Historical data (reference)
Autoencoder (CVAE)
Current pipeline (XGBoost + wind resampling)
Bars show variation BETWEEN annual averages, not hourly error or training loss.
Left: annual mean wind CF. Right: annual mean solar CF + wind CF.
Reference: 5 historical years (2020–2024); each generator: 15 annual profiles.
Retrospective pre-calibrated corpus; descriptive comparison, not a tail-probability estimate.